

YOUSIF ALDAKOKI

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Education

York University – Lassonde School of Engineering
Bachelor of Engineering, Software Engineering

Expected Apr. 2028
GPA: 8.60/9.00

Experience

Undergraduate Researcher

May 2025 – Aug. 2026

York University – research collaboration with University of Toronto and DRDC

- Led research ideation, method development, and experimental evaluation, resulting in **three accepted first-author papers at IEEE TNNLS, RadarConf, and GLOBECOM**.
- For **IQ-CMAE**, developed layer-specific contrastive gradient stopping to reduce interference with reconstruction learning; the model improved one-shot classification by **8.65 percentage points over a reconstruction-only baseline**.
- Reviewed **six manuscripts** across IEEE GLOBECOM and Open Journal of the Communications Society (OJCOMS).

Software Developer (Part-time) – DigitMakers, Clarica Group Inc.

Jun. 2024 – Present

- Deployed a Raspberry Pi pipeline streaming filament-extrusion PLC data to cloud dashboards, eliminating manual reporting for the line and enabling direct review of production data.
- Fixed Delta PLC motor/winder controls using ISPSOFT, ASDA-Soft, and CANopen, enabling **5 kg filament production** alongside 1 kg spools without machine-specific official support.
- Built an automated ad-creative pipeline and **scaled paid campaign budgets 3×** while maintaining **20×** **attributed return on ad spend (ROAS)**.
- Built a Shopify shipping integration proxy normalizing carrier rates, labels, and tracking APIs; designed and rebuilt web interfaces using TypeScript, React, and Remix.

Projects

Semantic Auto Linker – Obsidian Plugin

TypeScript, Transformers.js, Ollama

- Published an open-source plugin with **1,000+ downloads** in Obsidian's community directory; combines title and alias matching with local embedding retrieval to suggest links between notes.
- Built background vault analysis, embedding visualization, and a persistent review workflow for approving suggested links before modifying notes.

MLP From Scratch

C++, CUDA

- Implemented forward propagation, backpropagation, and weight updates without ML frameworks, achieving **97% MNIST test accuracy**; wrote CUDA kernels for accelerated matrix operations and inference.

Real-Time Waste Classification System

PyTorch, ONNX, Java

- Led software development for an ESP32-CAM and cloud-based ResNet-18 classifier with **94% held-out test accuracy**; won **1st place at York Engineering Competition** and represented York at Ontario Engineering Competition.

Selected Publications (Accepted, 2026)

IQ-CMAE: Multi-modal Contrastive Masked Autoencoder for IQ-based Signal Classification

IEEE Transactions on Neural Networks and Learning Systems (TNNLS) | [Code](#)

Transformer-based MCTS for Cognitive Radar Task Scheduling in Dynamic Environments

IEEE Radar Conference (RadarConf 2026)

Sparse Mixture-of-Experts Transformer for IQ-based 5G-NR Synchronization Signal Classification

IEEE Global Communications Conference (GLOBECOM 2026)

Technical Skills & Memberships

Languages: Python, C++, CUDA, Java, TypeScript, JavaScript

ML: PyTorch, PufferLib, ONNX, Transformers, Self-Supervised Learning (SSL), Reinforcement Learning (RL), Computer Vision (CV)

Development: React, Node.js, Remix, Git, Docker, AWS, Cloudflare Workers

Memberships: IEEE; IEEE Computational Intelligence Society (CIS)